

Mihai Stanciu

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EDUCATION

Brigham Young University

Provo, Utah

Master of Science in Mechanical Engineering, GPA 3.85/4.0

August 2026

Bachelor of Science in Mechanical Engineering, Minor in Mathematics, GPA 3.73/4.0

April 2024

SKILLS

- **Computer-Aided Design:** SOLIDWORKS (solids/surfaces, master/parametric modeling, sheet metal, assemblies), Ansys (structural, topology optimization), Onshape (Python API integration), Fusion 360, Eagle PCB
- **Programming Experience:** Python, C/C++, MATLAB, Arduino, LaTeX
- **Manufacturing/Prototyping Experience:** Additive Manufacturing, Mill, Lathe, Laser Cutting & Engraving, ARC Welding, PCB Milling & Populating, Plastics and Composites Molding
- **Languages:** French (proficient), Romanian (native)

ENGINEERING EXPERIENCE

Compliant Mechanisms and Robotics Research Assistant

June 2024 – Present

Department of Mechanical Engineering, Brigham Young University

Provo, Utah

- Redesigned the automated roller of an inflatable robotic truss through iterative CAD and prototyping, achieving 31% weight reduction, 51% cost savings, and doubled structural strength over the previous design.
- Engineered a multi-stage gear train using off-the-shelf components to increase torque output, enhancing efficiency and reliability in advancing the inflatable tube through the roller system.
- Constructed 21 roller units by selecting optimal manufacturing processes for each component, developing custom vise fixtures for precise manual machining, and maintaining tight tolerances to maximize mechanical efficiency and repeatability.
- Presented the inflatable robotic truss to NASA planetary scientists and aerospace engineers at the 2024 Big Idea Challenge Forum and Lunar Surface Innovation Consortium in Las Vegas, Nevada.

Mechanical Subsystem Team Leader

September 2023 – June 2024

BYU Mars Rover, Brigham Young University

Provo, Utah

- Led two teams of six engineering students in the design and integration of the rover's wheel, serial manipulator, elevator, and antenna subsystems, improving overall mechanical architecture and system reliability.
- Developed testing procedures to quantify mechanical play and joint uncertainty in the serial manipulator of the rover, leading to significant reductions in backlash and improved precision of the manipulator.
- Competed in the 2024 International University Rover Challenge, achieving the highest score among all teams in two of four competition tasks and placing 3rd overall among 102 universities worldwide.

Robotics and Dynamics Research Assistant

March 2022 – September 2023

Department of Mechanical Engineering, Brigham Young University

Provo, Utah

- Engineered components for the in-house manufacturing of four pneumatic continuum serial manipulators, selecting materials based on properties relevant to dynamic modeling.
- Performed tensile testing and stress-strain analysis to select the optimal spine material for 12 continuum joints.
- Assembled and integrated pneumatic continuum joints for four serial manipulators, supporting fabrication and experimental validation of the robotic systems.

Senior Technician

September 2021 – April 2024

Experiential Learning Center, Brigham Young University

Provo, Utah

- Designed and fabricated the enclosure for a portable gaming console using additive and advanced manufacturing techniques, overseeing mass production of 120 assembly kits.
- Created a complete design package in SOLIDWORKS, including part and assembly drawings and a bill of materials, enabling efficient bulk production of a heavy-duty worktable.
- Trained six new technicians to safely operate PCB milling equipment, 3D printers, a table saw, lathe, and mill.

INTERESTS

Skiing • Camping • Traveling • Astronomy • Sketching • Woodworking • Military History • Furniture Restoration